

Model Card – Automated Loan Approval Model (v1.0)

Model Details

Field	Description
Model name	Automated Loan Approval Model (ALAM)
Version	1.0 (May 2025 prototype)
Developed by	Janani Sri Sree Ganesh
Algorithm	Logistic Regression Classification
Framework / Libraries	Python 3.11, scikit-learn 1.4.0, pandas 2.2.2

Intended Use

- **Primary purpose:** Screen consumer-loan applications to estimate the probability of repayment (P) and recommend *auto-accept*, *auto-reject*, or *manual-review*.
- **Decision workflow:**
 - $P \leq 0.20 \rightarrow \textit{Auto-reject}$
 - $P \geq 0.80 \rightarrow \textit{Auto-accept}$
 - $0.20 < P < 0.80 \rightarrow$ Loan officer will manually review

Training & Evaluation Data

Attribute	Description
Source file(s)	<code>loan_dataset.csv</code> (funded loans)
Observations	50 000 approved loans disbursed Jan 2020–Dec 2022
Label	<code>loan_status</code> $\in \{\text{repaid}, \text{default}\}$
Temporal split	Train: Jan 2020–Jun 2022 (70%) Test: Jul 2022–Dec 2022 (30%)
Sampling	Chronological; no synthetic re-sampling (class weights applied)
Imbalance	Repayment rate 62.4% (class weight 1.66:1)
Known anomalies	5 694 rows with <code>employment_length_years</code> $>$ (age–15), flagged
Protected attributes	Proxies only (zip code, education, home-ownership)

*NB: Dataset contains **selection bias**: only approved loans are present.*

Out-of-sample Performance of the Model

Metric	Value
ROC AUC	0.835
PR AUC	0.800
Overall accuracy	0.78
Precision ($PD \geq 0.80$)	0.77
Recall ($PD \geq 0.80$)	0.71
F1-score ($PD \geq 0.80$)	0.74

Table 1: Out-of-sample performance of the scaled Logistic Regression classifier (test set, July–December 2022).

Sub-group fairness (Δ selection rate across proxies) all < 2 p.p.

Ethical & Regulatory Considerations

- **EU AI Act classification**
- **Human oversight:** Manual-review queue
- **Bias mitigation:** Class-weighting, fairness audits
- **Data minimisation & GDPR:** No direct gender/race storage
- **Potential harms:** False negatives (credit denial), false positives (over-indebtedness)

Limitations

- Excludes rejected loans (selection bias)
- Limited to pandemic period; macro-drift possible
- Proxies for protected attributes may under-detect bias
- Linear log-odds assumption; non-linear effects under-captured